

INTERNSHIP REPORT

On

**Log Anomaly Detection Web Portal for System Security
Monitoring**

By

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CERTIFICATE

This is to certify that the internship report entitled " **Log Anomaly Detection Web Portal for System Security Monitoring** " is a Bonafide record of work carried out by **P. Keerthana [2023003450], B. Sandeep Reddy [2023003382], P. Poorna Chandra Rao [2023003434]**, submitted in partial fulfilment of requirement for the award of degree of **Bachelor of Technology in Computer Science and Engineering**.


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ABSTRACT

The Banking Security Monitoring System (SOC Portal) is a web-based application designed to simulate a Security Operations Center (SOC) used in banking systems to monitor login activities and detect suspicious behavior. The system focuses on analyzing login logs and identifying anomalies such as brute force attacks and unusual login attempts.

The application allows users to upload CSV files containing simulated banking log data. Using JavaScript, the system processes the data, applies rule-based anomaly detection, and categorizes user activity into normal, suspicious, or high-risk. The results are displayed through an interactive dashboard using charts and tables.

This project demonstrates how cybersecurity monitoring systems work in real-world banking environments and highlights the importance of detecting threats using data analysis and visualization techniques.

CHAPTER 1: INTRODUCTION

1.1 Overview

With the rapid growth of digital banking, monitoring user activity has become essential to prevent unauthorized access and cyber threats. Banks use Security Operations Centers (SOC) to track login behavior and detect anomalies in real time.

The Banking Security Monitoring System (SOC Portal) is designed to simulate such a system using web technologies. It analyzes login data and identifies suspicious patterns using rule-based logic.

1.2 Motivation

- Increasing cyber attacks in banking systems
- Need for monitoring login activities
- Lack of access to real banking datasets
- Need for a simple SOC simulation system

1.3 Objectives

- To develop a web-based SOC dashboard
- To analyze login data from CSV files
- To detect anomalies using rule-based logic
- To visualize login trends using charts

1.4 Limitations of Existing Systems

- Real banking systems are not accessible
- Complex SOC tools require advanced infrastructure
- Lack of simple learning-based SOC simulations

CHAPTER 2: SYSTEM DESIGN & METHODOLOGY

2.1 System Architecture

The system consists of three main components:

- CSV Data Input
- Data Processing & Anomaly Detection
- Visualization Dashboard

2.2 Methodology

The system follows these steps:

- User uploads CSV file containing login logs
- System reads file using JavaScript FileReader
- Extracts user, login attempts, and timestamps
- Applies rule-based anomaly detection
- Displays results in dashboard.

2.3 Anomaly Detection Logic

- Login attempts $\geq 3 \rightarrow$ Brute Force Attack
- Login attempts = 2 $\rightarrow$ Suspicious Login
- Login attempts = 1 $\rightarrow$ Normal

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 Software Requirements

- HTML
- CSS
- JavaScript
- Chart.js

3.2 Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4GB
- Browser: Chrome/Edge

CHAPTER 4: IMPLEMENTATION

4.1 Frontend Implementation

The user interface is developed using HTML and CSS. It provides a simple dashboard where users can upload CSV files and view results.

4.2 Data Processing

JavaScript is used to read CSV data and extract relevant information such as login attempts and timestamps.

4.3 Visualization

Chart.js is used to create:

- Activity trend graphs
- Severity distribution charts

CHAPTER 5: FEATURES IMPLEMENTED

- CSV file upload
- Log monitoring dashboard
- Anomaly detection
- Status display (Success / Failed)
- Graph visualization
- Severity distribution chart
- System health monitoring
- Live log simulation

CHAPTER 5: SYSTEM DESIGN & METHODOLOGY

5.1 System Architecture

The Banking Security Monitoring System (SOC Portal) is designed as a web-based application that simulates a Security Operations Center used in banking systems. The system consists of three main components:

1. **Input Module (CSV Upload)**
Allows users to upload log data in CSV format containing login activities.
2. **Processing Module (Anomaly Detection)**
Processes the uploaded data using JavaScript and applies rule-based logic to detect suspicious activities.
3. **Visualization Module (Dashboard)**
Displays processed results using tables and charts for easy monitoring and analysis.

5.2 Dataset & Input

The dataset used in this project is a **simulated banking log dataset obtained from Kaggle**. Since real banking data is confidential, this dataset is used to represent how login logs are stored in real systems.

The dataset includes:

- Username
- Login attempts
- Timestamp
- Login status (Success / Failed)

This data helps in analyzing user behavior and detecting anomalies.

5.3 Methodology

The system follows a step-by-step process for monitoring and detecting suspicious activities:

1. **CSV File Upload**

The user uploads a CSV file containing login log data.

2. **File Reading**

The system reads the file using JavaScript FileReader.

3. **Data Extraction**

Important fields such as username, login attempts, and timestamp are extracted.

4. **Data Processing**

The system groups login attempts for each user.

5. **Anomaly Detection**

Rule-based logic is applied to identify suspicious behavior.

6. **Result Display**

The results are shown in a dashboard using tables and charts.

5.5 Visualization

The system uses **Chart.js** to display data in graphical format. The following visualizations are included:

- **Line Chart** → Shows login activity trends over time
- **Bar Chart** → Displays number of login attempts per user
- **Pie Chart** → Represents severity distribution (Normal, Suspicious, Attack)

These visualizations help in easy understanding and monitoring of system activity.

5.6 System Workflow

The overall workflow of the system is as follows:

- User uploads CSV file
- System reads and processes data
- Applies anomaly detection rules
- Classifies login behavior
- Displays results in dashboard

5.7 Deployment

The Banking Security Monitoring System (SOC Portal) is deployed as a web-based application, allowing users to access the system through a browser without requiring any additional installation.

CHAPTER 6: IMPLEMENTATION

The implementation of the Banking Security Monitoring System (SOC Portal) is divided into three main modules: frontend interface, data processing, and visualization dashboard.

6.1 Frontend Implementation

The user interface is developed using **HTML and CSS**, providing a simple and interactive layout for users to upload log files and view results.

Features:

- File upload option for CSV logs
- Dashboard layout for displaying results
- Tables to show login records
- Clean and user-friendly interface

The design ensures easy navigation and smooth user interaction.

6.2 Data Processing

The core logic of the system is implemented using **JavaScript**.

Process:

1. The uploaded CSV file is read using the **FileReader API**
2. Data is parsed and converted into structured format
3. Important fields such as:
 - Username
 - Login attempts
 - Timestamp

- Login status
are extracted
4. The system groups login attempts based on users

6.3 Anomaly Detection Implementation

- The anomaly detection is implemented using **rule-based logic** in JavaScript.
- **Logic Applied:**
- If login attempts $\geq 3 \rightarrow$ Brute Force Attack
- If login attempts = 2 $\rightarrow$ Suspicious Login
- If login attempts = 1 $\rightarrow$ Normal Activity
- The system categorizes each user activity based on these rules and assigns severity levels.

CHAPTER 6: RESULTS & ANALYSIS

The system successfully detects suspicious login activities based on predefined rules.

- Brute force attacks are identified accurately
- Suspicious logins are highlighted
- Dashboard provides clear visualization
- Charts help analyze login patterns

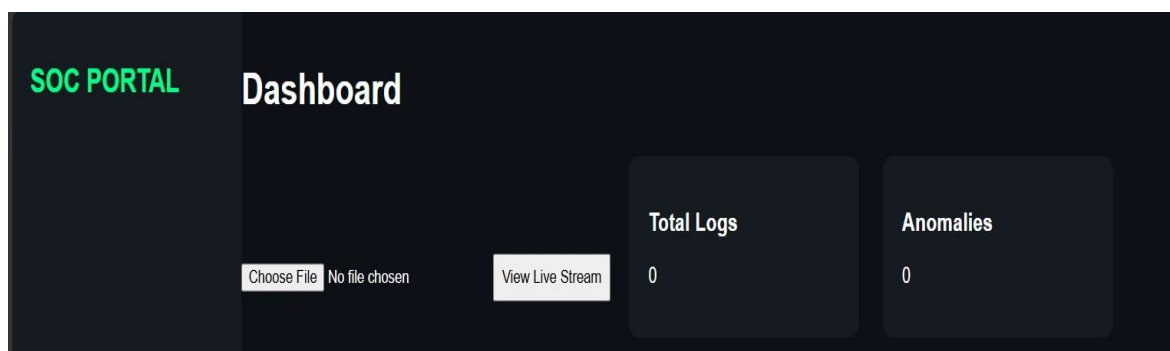
The system demonstrates how real SOC dashboards function in monitoring banking systems.

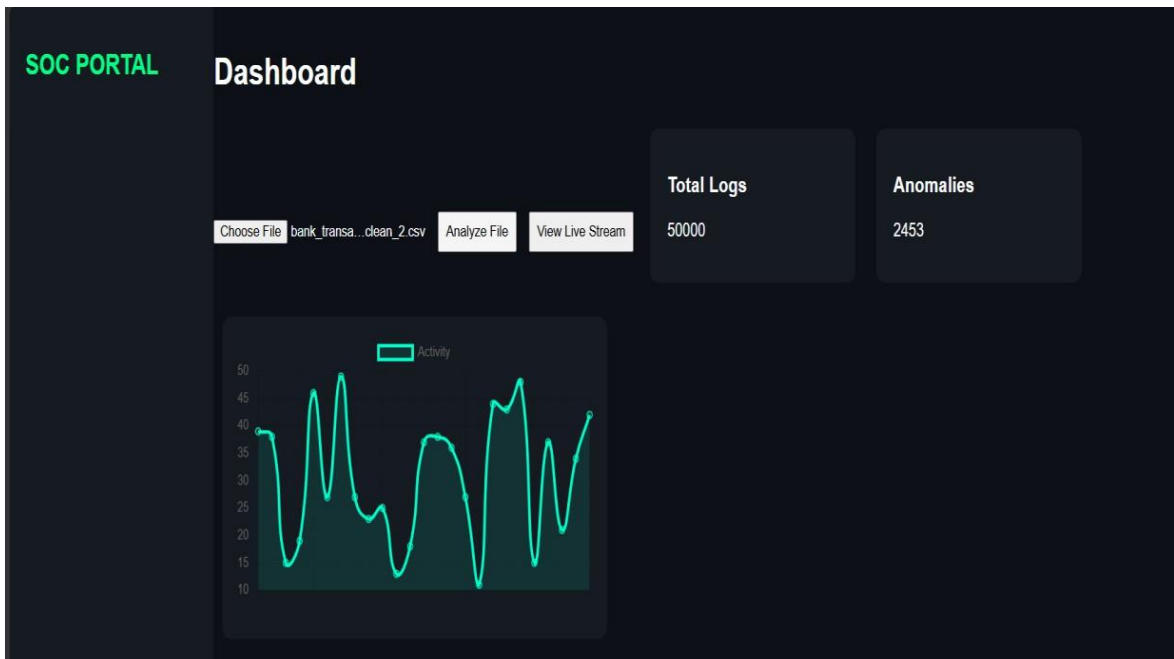
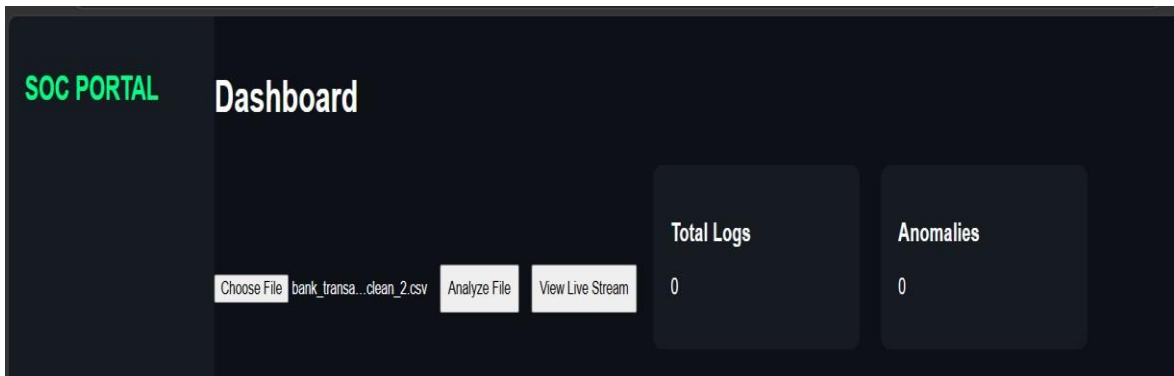
CHAPTER 7: LIMITATIONS

- No real-time banking integration
- Uses simulated dataset
- No backend/database
- Rule-based detection only

CHAPTER 8: FUTURE ENHANCEMENTS

- Integration with real-time APIs
- Machine Learning-based detection
- Backend database support
- Real-time alerts and notifications





System Health

-  Auth Service
-  API Gateway
-  DB Primary
-  Firewall
-  Log Collector
-  IAM Service

SOC PORTAL	10/30/2023 16:52	AC00317	1	Success	-
	12/14/2023 18:32	AC00359	1	Success	-
	2/21/2023 16:56	AC00242	1	Success	-
	6/8/2023 18:18	AC00285	1	Success	-
	2/28/2023 16:36	AC00002	1	Success	-
	8/11/2023 18:31	AC00014	1	Success	-
	6/6/2023 18:54	AC00095	1	Success	-
	5/2/2023 18:25	AC00453	3	Failed	⚠ Brute Force Attack
	3/2/2023 16:43	AC00241	1	Success	-
	10/20/2023 17:53	AC00041	1	Success	-
	4/17/2023 16:37	AC00441	5	Failed	⚠ Brute Force Attack
	3/29/2023 16:42	AC00057	1	Success	-
	1/24/2023 16:18	AC00390	1	Success	-
	9/20/2023 16:55	AC00313	1	Success	-
	7/17/2023 17:49	AC00367	1	Success	-
	2/6/2023 16:24	AC00291	1	Success	-

CHAPTER 9: CONCLUSION

The Banking Security Monitoring System (SOC Portal) successfully simulates a SOC dashboard used in banking systems. It demonstrates how login activities can be monitored and analyzed to detect suspicious behavior.

The project highlights the importance of cybersecurity and provides a foundation for building advanced security monitoring systems.

CHAPTER 10: REFERENCES

- Kaggle Dataset (Banking Logs)
Available at: <https://www.kaggle.com>
- Chart.js Documentation
- HTML, CSS, JavaScript Documentation